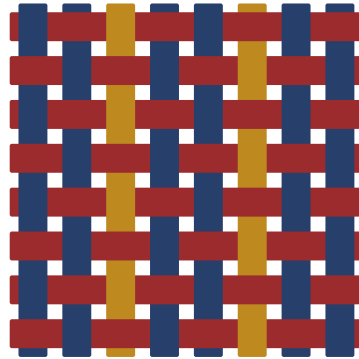




The Singular Value Decomposition

One decomposition, every question · fill-in notes

woven on Loom

June 28, 2026

dyed in indigo, madder & weld; ruled in iron-gall. woven on Loom.

THE ONE IDEA, AND HOW TO READ THESE NOTES

□□□□

Every real matrix A factors as

$$A = U \Sigma V^T, \quad U, V \text{ orthogonal}, \quad \Sigma = \text{diag}(\sigma_1 \geq \sigma_2 \geq \dots \geq 0).$$

The numbers σ_i are the **singular values**. The single idea of these notes: **ask a matrix almost any question, and the answer is read straight off its singular values**. Rank, size, invertibility, the best low-rank picture, the four subspaces, the condition number: all of them live in Σ .

The master table: read it off Σ (fill it in as you go)

Question about A	Answer, read off Σ
rank of A	number of _____ singular values = r (§3)
operator norm $\ A\ _2$	σ _____ (the largest)
Frobenius norm $\ A\ _F$	$\sqrt{\sigma_1^2 + \dots + \sigma_r^2}$
is A (square) invertible?	yes iff σ_n _____ 0
condition number $\kappa(A)$	σ_1/σ_r (§5)
best rank- k approximation	keep $\sigma_1, \dots, \sigma_k$, drop the rest (§4)
$ \det A $ (square)	$\sigma_1 \sigma_2 \dots \sigma_n$
the four fundamental subspaces	spans of columns of U and V (§3)

How to read (passive *and* active)

- **Read** the strands and the stated theorems.
- **Fill** the blanks _____, the [fill in: missing step] proof gaps, and the *Your turn* boxes. Your linear-algebra course is the answer key.
- Bump each section's \warmth{0} to 0–5 once you can rebuild it cold.

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1 THE SHAPE OF A MATRIX

□□□□ Every linear map is a rotation, a stretch along axes, and another rotation.

Forget the entries for a moment and watch what $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ does. Feed it the unit sphere; out comes an **ellipsoid**. The singular value decomposition is just the honest description of that ellipsoid: the directions of its axes, and how long they are. Reading $A = U\Sigma V^T$ right to left, A is a **rotation** V^T , then an axis-aligned **stretch** Σ , then another rotation U . Three simple motions, in that order, are *all* any linear map ever does.

The decomposition

Definition 1.1 (singular value decomposition). A **singular value decomposition** of $A \in \mathbb{R}^{m \times n}$ is a factorization $A = U\Sigma V^T$ where

- $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are **orthogonal**: $U^T U = \underline{\hspace{2cm}}$ and $V^T V = \underline{\hspace{2cm}}$;
- $\Sigma \in \mathbb{R}^{m \times n}$ is "diagonal" with entries $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 = \sigma_{r+1} = \dots$, the **singular values**.

The columns u_i of U are the **left** singular vectors; the columns v_i of V the **right** singular vectors.

The singular triple

Columns of $A = U\Sigma V^T$ say it most vividly. Each triple (σ_i, u_i, v_i) satisfies

$$A v_i = \sigma_i u_i \quad \text{and} \quad A^T u_i = \sigma_i v_i.$$

So A takes the orthonormal directions v_i to the orthogonal directions $\sigma_i u_i$: the v_i are the *axes of the input circle*, the $\sigma_i u_i$ the *axes of the output ellipse*.

Your turn. A maps the unit sphere to an ellipsoid whose semi-axis lengths are the singular values.

- For $A = \text{diag}(3, 2)$ the unit circle becomes an ellipse with semi-axes $\underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}}$, along the $\underline{\hspace{2cm}}$ directions.
- If some $\sigma_i = 0$, the ellipsoid is *flat* in that direction. Sketch what the unit circle maps to when $A = \text{diag}(2, 0)$.

Why insist on the ordering $\sigma_1 \geq \sigma_2 \geq \dots$? Because then the *first* singular triple is the single most important thing A does: σ_1 is how much A can stretch any unit vector, achieved in direction v_1 . Every later triple is the best that remains after the earlier ones are used up. This "greedy from the top" structure is the whole reason truncating the SVD gives the best simple picture (§4).

? Orthogonal matrices are exactly the rigid motions fixing the origin: they preserve every length and angle. $\det = \pm 1$ (rotation or reflection).

? A flat ellipse = a line segment. That collapse is exactly rank loss; see §3.

2 WHY IT ALWAYS EXISTS

□□□□ Eigen-decomposition can fail; the SVD never does. The reason is $A^T A$.

Eigenvalues are fussy: a matrix may not be diagonalizable, may have complex eigenvalues, must be square. The SVD has *none* of these defects. The trick is to launder A through the one matrix that is always perfectly behaved: $A^T A$ is symmetric and positive semidefinite, so the **spectral theorem** hands us an orthonormal eigenbasis on a silver platter. The right singular vectors are those eigenvectors; the singular values are square roots of the eigenvalues.

Existence, in one construction

Theorem 2.1 (every matrix has an SVD). Every $A \in \mathbb{R}^{m \times n}$ admits a singular value decomposition $A = U \Sigma V^T$.

Proof. Skeleton (carry it out as you read):

1. $A^T A$ is symmetric PSD, so by the spectral theorem $A^T A = V \Lambda V^T$ with V orthonormal and $\Lambda = \text{diag}(\lambda_1 \geq \dots \geq \lambda_n \geq 0)$.
2. Set $\sigma_i = \sqrt{\lambda_i}$. For each i with $\sigma_i > 0$, define $u_i = \frac{1}{\sigma_i} A v_i$. [fill in: check $\{u_i\}$ is orthonormal, using $v_i^T A^T A v_j = \lambda_j v_i^T v_j$]
3. Extend $\{u_i\}$ to an orthonormal basis U of $\mathbb{R}^m$. [fill in: verify $A = \sum_i \sigma_i u_i v_i^T = U \Sigma V^T$ by checking both sides agree on the basis $\{v_j\}$]

Remark 2.2 (how unique?). The singular values σ_i are unique. The singular vectors are not: each (u_i, v_i) may be flipped in sign together, and within a repeated singular value you may rotate the corresponding vectors. So Σ is rigid; U, V have a little freedom.

Your turn. Compute the SVD of $A = \begin{pmatrix} 0 & 2 \\ 3 & 0 \end{pmatrix}$ by the recipe above.

$$A^T A = \underline{\hspace{2cm}}, \quad \text{eigenvalues } \underline{\hspace{2cm}} \Rightarrow \sigma_1, \sigma_2 = \underline{\hspace{2cm}},$$

$$v_1 = \underline{\hspace{2cm}}, \quad v_2 = \underline{\hspace{2cm}}, \quad u_i = \frac{1}{\sigma_i} A v_i : \quad u_1 = \underline{\hspace{2cm}}, \quad u_2 = \underline{\hspace{2cm}}.$$

When A is symmetric PSD, the SVD and the eigendecomposition coincide ($U = V$, $\sigma_i = \lambda_i$). For symmetric but indefinite A , $\sigma_i = |\lambda_i|$ and a sign moves into U . The SVD is the eigendecomposition's well-behaved cousin.

? Mirror route: $AA^T = U \Sigma \Sigma^T U^T$ gives the left vectors u_i directly. $A^T A$ and AA^T share the nonzero eigenvalues σ_i^2 .

? $A^T A = \text{diag}(9, 4)$, so $\sigma = 3, 2$, $v_i = e_i$ and $u_1 = e_2, u_2 = e_1$. Notice $U \neq V$: this map is a reflection-and-stretch, not a symmetric one.

3 THE FOUR SUBSPACES, IN ONE BREATH

□□□□ U splits the output space, V splits the input space, along the rank.

Cut the singular vectors at the rank r (the number of nonzero σ_i). The first r columns of V and the last $n - r$ split the input $\mathbb{R}^n$ into two orthogonal pieces; the first r columns of U and the last $m - r$ split the output $\mathbb{R}^m$. Those four pieces are exactly the **four fundamental subspaces** of A , and the SVD names all of them at once.

Theorem 3.1 (*the four subspaces*). Let $A = U\Sigma V^T$ have rank r (so $\sigma_1, \dots, \sigma_r > 0$ and the rest vanish). Then:

subspace	lives in	is the span of ... (dimension)
column space $\text{col}(A)$	$\mathbb{R}^m$	$u_1, \dots, u_r$ (dim = r)
left null space $\text{null}(A^T)$	$\mathbb{R}^m$	$u_{r+1}, \dots, u_m$ (dim = $m - r$)
row space $\text{col}(A^T)$	$\mathbb{R}^n$	$v_1, \dots, v_r$ (dim = r)
null space $\text{null}(A)$	$\mathbb{R}^n$	$v_{r+1}, \dots, v_n$ (dim = $n - r$)

The picture behind the table: A sends the row space *isomorphically* onto the column space, stretching the i -th axis by σ_i , and it *annihilates* the null space. Nothing else happens. Rank-nullity ($\dim \text{null}(A) = n - r$) and the orthogonal splittings $\text{col}(A) \perp \text{null}(A^T)$ in $\mathbb{R}^m$, $\text{col}(A^T) \perp \text{null}(A)$ in $\mathbb{R}^n$ are now visibly true, because the columns of an orthogonal matrix are orthonormal.

Your turn. Take $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, with SVD given by $\sigma_1 = 2$, $\sigma_2 = 0$, $u_1 = v_1 = \frac{1}{\sqrt{2}}(1, 1)$, $v_2 = \frac{1}{\sqrt{2}}(1, -1)$. Read off:

$\text{rank}(A) = \underline{\hspace{1cm}}$, $\text{col}(A) = \text{span } \underline{\hspace{2cm}}$, $\text{null}(A) = \text{span } \underline{\hspace{2cm}}$.

Pseudoinverse preview: because A is a clean isomorphism row space $\rightarrow$ column space, it has an obvious inverse *there* (divide by σ_i) and zero elsewhere. That is the Moore--Penrose pseudoinverse of §5.

? Rank, finally honest: $\text{rank}(A)$ = the number of nonzero singular values. Tiny-but-nonzero σ_i are why "numerical rank" is a threshold on Σ , not a yes/no. = $\text{span}\{(1, 1)\}$; null space = $\text{span}\{(1, -1)\}$. The map flattens the plane onto a line.

4 THE BEST SIMPLE PICTURE

□□□□ Keep the top few singular triples and you have the best low-rank copy of A . Provably.

Here is why the SVD is everywhere in data science. Write A as a sum of rank-one layers,

$$A = \sum_{i=1}^r \sigma_i u_i v_i^\top,$$

ordered by importance $\sigma_1 \geq \sigma_2 \geq \dots$. Keep only the top k layers and you get $A_k = \sum_{i \leq k} \sigma_i u_i v_i^\top$, the **truncated SVD**. The miracle: among *all* matrices of rank $\leq k$, A_k is the closest one to A . Compression, denoising, PCA, recommender systems, are all this one theorem.

Theorem 4.1 (Eckart–Young–Mirsky). For every B with $\text{rank}(B) \leq k$,

$$\|A - B\| \geq \|A - A_k\|,$$

in both the operator and the Frobenius norm. The optimal error is read off the discarded singular values:

$$\|A - A_k\|_2 = \sigma_{k+1}, \quad \|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \dots + \sigma_r^2}.$$

Proof. Two halves:

1. *The error of A_k .* $A - A_k = \sum_{i>k} \sigma_i u_i v_i^\top$ is itself in SVD form, so its singular values are $\sigma_{k+1}, \sigma_{k+2}, \dots$ [fill in: read $\|\cdot\|_2$ and $\|\cdot\|_F$ off those]
2. *Nobody does better.* Let $\text{rank}(B) \leq k$, so $\text{null}(B)$ has dimension $\geq n-k$. It meets $\text{span}\{v_1, \dots, v_{k+1}\}$ (dimension $k+1$) in some unit vector w . [fill in: show $Bw = 0$ and $\|Aw\| \geq \sigma_{k+1}$, hence $\|A - B\|_2 \geq \|(A - B)w\| = \|Aw\| \geq \sigma_{k+1}$]

The reason this works is the ordering. Total “energy” $\|A\|_F^2 = \sum_i \sigma_i^2$ piles up in the largest few singular values, so throwing away the small ones costs almost nothing. A picture, a covariance matrix, a ratings table: each is usually *approximately* low rank, meaning its σ_i decay fast, and the truncated SVD is a faithful thumbnail at a fraction of the storage.

Your turn. Suppose A (4×4) has singular values $\sigma = (10, 6, 2, 1)$. For the rank-2 truncation A_2 :

$$\|A - A_2\|_2 = \underline{\hspace{2cm}}, \quad \|A - A_2\|_F = \underline{\hspace{2cm}}, \quad \frac{\|A_2\|_F^2}{\|A\|_F^2} = \underline{\hspace{2cm}}.$$

Caveat worth knowing: Eckart–Young is for *unitarily invariant* norms (operator, Frobenius, all Schatten- p). For other measures (entrywise ℓ_1 , nonnegative or sparse factors), best low-rank approximation can be NP-hard. The SVD’s optimality is special, not universal.

? Spectral error
 $= \sigma_3 = 2$;
 Frobenius error
 $= \sqrt{2^2 + 1^2} = \sqrt{5}$;
 energy kept
 $= \frac{10^2 + 6^2}{10^2 + 6^2 + 2^2 + 1^2} =$
 $\frac{136}{141} \approx 96.5\%$. Two
 numbers carry
 the matrix.

5 FACES OF THE SVD

□□□□ One factorization, a dozen classical tools. Each is a small edit to Σ .

With $A = U\Sigma V^T$ in hand, a surprising number of textbook objects are just Σ , lightly edited. Invert the nonzero singular values and you get the pseudoinverse; take their ratio and you get the condition number; throw Σ away entirely and keep UV^T and you get the nearest orthogonal matrix. Here is the keyring.

The keyring: each tool is Σ , edited

tool	formula	what it does
pseudoinverse A^+	$V\Sigma^+U^T$	least squares: $x^* = A^+b$ minimizes $\ Ax - b\ $
condition number $\kappa_2(A)$	$\frac{\sigma_1}{\sigma_r}$	how much solving $Ax = b$ amplifies error
polar factor Q	UV^T	the orthogonal matrix <i>nearest</i> to A
PCA directions	top columns of V	principal axes; $\sigma_i^2 \propto$ variance captured
operator / Frobenius norm	$\sigma_1 / \sqrt{\sum \sigma_i^2}$	size of A as a map / as a vector
best rank- k copy	truncate Σ to k	§4

Definition 5.1 (Moore–Penrose pseudoinverse). $A^+ = V\Sigma^+U^T$, where Σ^+ replaces each nonzero σ_i by $\frac{1}{\sigma_i}$ and keeps the zeros (and transposes the shape). It is the honest inverse on the part of the space A handles cleanly: it inverts the stretch on the row space and sends the rest to 0.

Your turn. Least squares in one line. To minimize $\|Ax - b\|_2$ over x , project b onto the column space and undo the stretch:

$$x^* = A^+b = \sum_{\sigma_i > 0} \frac{u_i^T b}{\sigma_i} v_i.$$

Where does a small σ_i make this dangerous, and which face in the keyring measures exactly that danger? _____

The one idea, recovered

Look back at the master table on the first page and check that you can now fill every row. A single factorization, $A = U\Sigma V^T$, answered all of it: the rank is the count of nonzero σ_i , the norms and condition number are simple functions of them, the four subspaces are the columns of U and V cut at the rank, the best low-rank copy keeps the top few, and the pseudoinverse inverts what can be inverted. **Ask a matrix a question; read the answer off its singular values.**

? When A is square and invertible, $\Sigma^+ = \Sigma^{-1}$ and $A^+ = A^{-1}$. The pseudoinverse only earns its keep when A is rectangular or rank-deficient.

? A tiny σ_i blows up $u_i^T b / \sigma_i$ – the condition number $\kappa_2 = \sigma_1 / \sigma_r$ is the size of that hazard. Regularization (ridge / truncated SVD) tames it.

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Where to go next: the SVD is to general matrices what the spectral theorem is to symmetric ones. For *huge*  $A$  you never form  $A^T A$ ; randomized SVD and Krylov methods get